Classical Field Theory — Exercise Sheet 18

Spectrum and moving kinks

Solutions

Retain $b(x) = \tanh x$, $V(s) = \frac{1}{2}(1-s^2)^2$ and $H_K = -\partial_x^2 + 4 - 6\operatorname{sech}^2 x$ with domain $H^2 \subset L^2$. Lecture 18 proves $H_K = A^\dagger A$ for $A = \partial_x + 2\tanh x$ and $\ker H_K = \operatorname{span}\{\operatorname{sech}^2 x\}$. Use its H^1 form $\mathfrak{q}[h] = \|Ah\|_2^2$ and density/integration-by-parts results.

Exercise 18.1 (The full spectrum through two adjoint pairs). The following analytic spectral theorem is given for this problem: if W is real, bounded and smooth, with $W(x) \to 0$ at both ends, then $-\partial_x^2 + 4 + W$ on $H^2(\mathbb{R})$ is self-adjoint, its essential spectrum is $[4, \infty)$, and every spectral point outside that interval is an isolated finite-multiplicity eigenvalue. This follows from Teschl, *Mathematical Methods in Quantum Mechanics* (2009), Lemmas 9.33 and 9.35, pp. 215–216, after shifting by four; the discrete/essential spectrum convention is in §6.4, p. 145. This input supplies more than the formal factorisation in Lecture 18. Set $B = \partial_x + \tanh x$ with domain H^1.

(a) Find $B^\dagger$ and the product domains, and prove $AA^\dagger = B^\dagger B + 3$ and $BB^\dagger + 3 = -\partial_x^2 + 4$. Use these identities to classify all L^2 eigenvalues of H_K, including possible embedded or threshold eigenvalues. Check the domains when intertwining.

(b) Normalise its two bound states. Exhibit a bounded solution at spectral value four and explain why it is not another L^2 eigenstate. Using the given theorem, state the full spectrum.

Solution. For the first part, compact integration by parts gives $B^\dagger = -\partial_x + \tanh x$ on H^1. Conversely a vector in the adjoint domain has distributional derivative in L^2, because the coefficient is bounded, so that is the full domain. Multiplication by $\tanh x$ preserves H^1. As in Lecture 18, $u \in H^1$ and $Bu \in H^1$ are equivalent to $u \in H^2$; the same argument applies to $B^\dagger$ and $A^\dagger$. All product domains in the following identities are therefore H^2:

$$AA^\dagger = -\partial_x^2 + 4 - 2\operatorname{sech}^2 x = B^\dagger B + 3, \qquad BB^\dagger + 3 = -\partial_x^2 + 4 =: H_0.$$

The identities first follow by product differentiation on compact tests, then extend by H^2 density.

Suppose $H_K u = \lambda u$ with $u \in H^2$ nonzero. Self-adjointness makes λ real, and factorisation makes it nonnegative. The zero case is the known one-dimensional kernel. The equation $u'' = (4 - 6\operatorname{sech}^2 x - \lambda)u$ and the bounded derivatives of its coefficient bootstrap u into every integer Sobolev space: Leibniz' rule makes multiplication by this coefficient bounded on each integer H^s; if $u \in H^s$, its right side is in H^s, so Fourier transformation gives $u \in H^{s+2}$. Products and first derivatives in the following intertwinings thus belong to the actual operator domains. For $\lambda > 0$, $v = Au$ is nonzero and

$$(B^\dagger B + 3)v = \lambda v.$$

If $Bv \neq 0$, then $z = Bv \in H^2 \cap L^2$ obeys $H_0 z = \lambda z$. Fourier transformation gives $(k^2 + 4 - \lambda)\widehat{z} = 0$. The zero set of this multiplier is empty or finite, hence has measure zero; no nonzero L^2 Fourier profile can be supported there. Thus $Bv = 0$. Solving the first-order equation gives $v = C\operatorname{sech} x$, and then necessarily $\lambda = 3$. The map A is injective on a positive eigenspace, so

this candidate has multiplicity at most one. This excludes all other eigenvalues, including those in $[4,\infty)$.

For the second part, $A^\dagger \operatorname{sech} x = 3\operatorname{sech} x \tanh x$ constructs a nonzero eigenfunction at three. The substitution $z = \tanh x$ gives

$$\int \operatorname{sech}^4 x\,\mathrm{d}x = \frac{4}{3}, \qquad \int \operatorname{sech}^2 x \tanh^2 x\,\mathrm{d}x = \frac{2}{3}.$$

Thus the positive normalisations are

$$\psi_0 = \frac{\sqrt{3}}{2}\operatorname{sech}^2 x, \qquad \psi_3 = \sqrt{\frac{3}{2}}\operatorname{sech} x \tanh x.$$

They are orthogonal by parity. Applying the formal differential expressions of $A^\dagger B^\dagger$ to the free threshold solution one gives $3\tanh^2 x - 1$; this step does not apply the L^2 operators to the non-L^2 constant. Direct substitution verifies that the resulting smooth profile solves the differential equation $H_K u = 4u$. Half this function tends to one at both ends, so it is bounded but not L^2. Finally the given theorem supplies essential spectrum $[4,\infty)$ and allows only isolated eigenvalues elsewhere. The classification above proves

$$\sigma(H_K) = \{0,3\} \cup [4,\infty).$$

No claim about nonlinear stability follows from this spectral calculation.

Exercise 18.2 (A moving kink and the sign of momentum). For $|v_{\mathrm{tr}}| < 1$, set $\gamma = (1-v_{\mathrm{tr}}^2)^{-1/2}$ and

$$\phi(t,x) = \tanh\bigl(\gamma(x - v_{\mathrm{tr}}t - x_0)\bigr).$$

Verify the nonlinear field equation directly. At fixed time compute the charge, full energy and the lower-index translation integral $P_1 = \int \phi_t \phi_x \mathrm{d}x$. Find the physical raised momentum P^1 and the relation between its square and the energy square. Check the result at $v_{\mathrm{tr}} = 3/5$. Justify membership in the affine class $\phi - b \in H^1$, $\phi_t \in L^2$.

Solution. Write $\xi = \gamma(x - v_{\mathrm{tr}}t - x_0)$. The field equation becomes

$$\phi_{tt} - \phi_{xx} + V'(\phi) = \bigl[\gamma^2(v_{\mathrm{tr}}^2 - 1) + 1\bigr] b''(\xi) = 0.$$

At fixed time the spatial endpoint limits remain -1 and 1, so the charge is one. The derivatives and potential are

$$\phi_x = \gamma \operatorname{sech}^2 \xi, \qquad \phi_t = -\gamma v_{\mathrm{tr}} \operatorname{sech}^2 \xi, \qquad V(\phi) = \frac{1}{2}\operatorname{sech}^4 \xi.$$

Since $\mathrm{d}x = \mathrm{d}\xi/\gamma$ and $\int \operatorname{sech}^4 \xi \mathrm{d}\xi = 4/3$,

$$E = \frac{\gamma^2(1+v_{\mathrm{tr}}^2)+1}{2\gamma}\frac{4}{3} = \frac{4}{3}\gamma, \qquad P_1 = -\gamma v_{\mathrm{tr}}\frac{4}{3}.$$

Lecture 3 gives $P^1 = -P_1$, whence $E^2 - (P^1)^2 = 16\gamma^2(1-v_{\mathrm{tr}}^2)/9 = 16/9$. For $v_{\mathrm{tr}} = 3/5$, $\gamma = 5/4$ and the values are $E = 5/3$, $P_1 = -1$, $P^1 = 1$. Both the boosted profile and b approach the same vacua exponentially at each fixed time. Their difference and its derivative therefore decay exponentially, as does ϕ_t. This proves the stated affine Sobolev membership.

Exercise 18.3 (Removing the linear translation drift). Use the spectral theorem for the self-adjoint operator in Problem 1 as an additional analytic input: on the orthogonal complement of its zero mode, the form-domain inequality is $\mathfrak{q}[h] \geq 3\|h\|_2^2$ for $h \in H^1$ with $(\psi_0, h) = 0$. Prove $\|h\|_{H^1}^2 \leq 2\mathfrak{q}[h]$ on this subspace. For any real solution of $h_{tt} + H_K h = 0$ with

$$h \in C(H^2) \cap C^1(H^1) \cap C^2(L^2),$$

prove that $(\psi_0, h(0)) = (\psi_0, h_t(0)) = 0$ preserves this orthogonality, and deduce a uniform bound on $\|h(t)\|_{H^1}^2 + \|h_t(t)\|_2^2$ from its initial quadratic energy. No existence theorem for this new linear equation is required: the assertion is conditional on a solution in the displayed class.

Solution. The form identity and the coefficient bound give

$$\|h_x\|_2^2 \leq \mathfrak{q}[h] + 2\|h\|_2^2, \qquad \|h\|_{H^1}^2 \leq \mathfrak{q}[h] + 3\|h\|_2^2 \leq 2\mathfrak{q}[h].$$

For the stated solution regularity, $c(t) = (\psi_0, h(t))$ is twice continuously differentiable and self-adjointness gives $c'' = -(\psi_0, H_K h) = -(H_K \psi_0, h) = 0$. Both initial values vanish, hence $c(t) = 0$ throughout the interval. The actual H^2 domain and the H^1 time derivative justify integration by parts when differentiating the quadratic form. It gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{E}_{\mathrm{lin}} = (h_t, h_{tt} + H_K h) = 0.$$

Applying the form bound at each time yields

$$\|h(t)\|_{H^1}^2 + \|h_t(t)\|_2^2 \leq 2\mathfrak{q}[h(t)] + \|h_t(t)\|_2^2 \leq 4\mathcal{E}_{\mathrm{lin}}(0).$$

Nonlinear evolution does not satisfy this projected linear equation. A nonlinear moving centre and its orthogonality condition would require additional arguments; the estimate itself establishes only the stated linear control.

Exercise 18.4 (The nonlinear energy remainder on a fixed slice). For real $h \in H^1(\mathbb{R})$, compute the exact remainder in $E_{\mathrm{stat}}[b+h] - E_{\mathrm{stat}}[b] - \mathfrak{q}[h]/2$. Prove a bound using only $\|h\|_{H^1}$. If $(\psi_0, h) = 0$, show that for $\|h\|_{H^1} \leq 1/(16\sqrt{2})$,

$$E_{\mathrm{stat}}[b+h] - E_{\mathrm{stat}}[b] \geq \frac{1}{8}\|h\|_{H^1}^2.$$

Why is this static estimate insufficient by itself to prove nonlinear orbital stability of the kink?

Solution. The polynomial expansion gives

$$V(b+h) = V(b) + V'(b)h + \tfrac{1}{2}V''(b)h^2 + 2bh^3 + \tfrac{1}{2}h^4.$$

The kinetic cross term and the linear potential term integrate to zero by $-b'' + V'(b) = 0$. This holds first for compact smooth perturbations, then for H^1 perturbations by density, since $b', b'', V'(b) \in L^2$. All cubic and quartic terms are integrable by the one-dimensional embedding. Thus the exact remainder is

$$\mathcal{R}[h] = \int (2bh^3 + \tfrac{1}{2}h^4)\mathrm{d}x.$$

Using $|b| \le 1$ and $\|h\|_\infty \le \|h\|_{H^1}/\sqrt{2}$ gives

$$|\mathcal{R}[h]| \le 2\|h\|_\infty \|h\|_2^2 + \tfrac{1}{2}\|h\|_\infty^2 \|h\|_2^2 \le \sqrt{2}\|h\|_{H^1}^3 + \tfrac{1}{4}\|h\|_{H^1}^4.$$

Under the orthogonality condition, $\mathfrak{q}[h]/2 \ge \|h\|_{H^1}^2/4$. Writing $\delta = 1/(16\sqrt{2})$, the remainder coefficient satisfies $\sqrt{2}\delta + \delta^2/4 = 1/16 + 1/2048 < 1/8$. Subtracting it proves the requested lower bound. This controls a fixed slice satisfying a fixed orthogonality condition. It neither selects a moving translation parameter nor proves that the nonlinear solution remains on such slices. Those are among the missing steps in an orbital-stability proof.